

Multiple Choice

1. **Answer:** A

Options: A) $10/e$; B) 10; C) $10e$; D) e^{10}

Note: The line $y = 10x$ is tangent to the curve $y = e^{(bx)}$ at some point in the xy -plane when their slopes are equal at that point, leading to the condition that the derivative of $e^{(bx)}$, which is $be^{(bx)}$, must equal 10, resulting in $b = 10/e$ when evaluated at the point where both functions intersect.

2. **Answer:** B

Options: A) 4; B) 6; C) 8; D) 9

Note: The correct answer is 6 because the number of structurally distinct Abelian groups of a given order can be determined by the partitioning of the prime factorization of that order, and for the order 72, which factors as $2^3 \times 3^2$, there are 6 distinct partitions corresponding to the 6 distinct Abelian group structures.

3. **Answer:** A

Options: A) $y = 2x + 1$; B) $y = 2x$; C) $y = x + 2$; D) $y = x + 1$

Note: The correct answer is $y = 2x + 1$ because the slope of the tangent line at $x = 0$ is the derivative of $y = x + e^x$ evaluated at that point, which is $1 + e^0 = 2$, and the y -intercept at that point is 1, giving the equation of the tangent line as $y = 2x + 1$.

4. **Answer:** C

Options: A) 4; B) 8; C) 12; D) 6

Note: The order of the element $(4, 2)$ in the group $Z_{12} \times Z_8$ is determined by the least common multiple of the orders of its components, with the order of 4 in Z_{12} being 3 and the order of 2 in Z_8 being 4, leading to the least common multiple of 3 and 4, which is 12.

5. **Answer:** B

Options: A) 0; B) 2; C) 1; D) 3

Note: The correct answer is 2 because for $Z_3[x]/(x^3 + cx^2 + 1)$ to be a field, the polynomial $x^3 + cx^2 + 1$ must be irreducible over Z_3 , which occurs when $c = 2$, as it does not have roots in Z_3 and cannot be factored into polynomials of lower degree.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because a group of order 102 can have a non-abelian structure, as it can be isomorphic to the symmetric group S_3 , which has order 6 and is non-abelian, and groups of order 102 can contain subgroups of such orders.

7. **Answer:** False

Options: A) True; B) False

Note: The order of the factor group $\mathbb{Z}_6/\langle 3 \rangle$ is actually 2, making the statement true, as the subgroup $\langle 3 \rangle$ consists of the elements $\{0, 3\}$ in $\mathbb{Z}_6$, leading to the factor group having the elements $\{\langle 0 \rangle, \langle 3 \rangle\}$, which confirms its order is indeed 2.

8. **Answer:** False

Options: A) True; B) False

Note: A complete space is not necessarily compact because completeness pertains to the convergence of Cauchy sequences, while compactness is related to the covering properties of the space, as demonstrated by the fact that the closed interval $[0, 1]$ is complete and compact, whereas the entire real line $\mathbb{R}$ is complete but not compact.

9. **Answer:** True

Options: A) True; B) False

Note: The set of all n th roots of unity under multiplication forms an abelian group because it satisfies the four group axioms-closure, associativity, identity, and invertibility-and additionally, multiplication of complex numbers is commutative, thereby ensuring the group is abelian.

10. **Answer:** True

Options: A) True; B) False

Note: The degree of the field extension $\mathbb{Q}(2 + 3i)$ over $\mathbb{Q}$ is 4 because the minimal polynomial of $2 + 3i$ over $\mathbb{Q}$ has degree 4, which corresponds to the product of the degrees of the minimal polynomials of 2 and 3 over $\mathbb{Q}$, each of which is 2.

Long Form Response

11. **Answer:** We can write $f(x) = \sqrt{5x^2 + 2x\sqrt{5} + 1} + x\sqrt{5}$

$$= \sqrt{(x\sqrt{5} + 1)^2} + x\sqrt{5}$$

$$= |x\sqrt{5} + 1| + x\sqrt{5}. \text{ If } x \leq -\frac{1}{\sqrt{5}}, \text{ then}$$

$$f(x) = |x\sqrt{5} + 1| + x\sqrt{5} = -x\sqrt{5} - 1 + x\sqrt{5} = -1.$$

$$\text{If } x \geq -\frac{1}{\sqrt{5}}, \text{ then } f(x) = |x\sqrt{5} + 1| + x\sqrt{5}$$

$$= x\sqrt{5} + 1 + x\sqrt{5}$$

$$= (x\sqrt{5} + 1) + (x\sqrt{5} + 1) - 1$$

$$\geq -1. \text{ Thus, the minimum value of } f(x) \text{ is } -1.$$

Note: The answer correctly identifies the minimum value of the function $f(x)$ by analyzing the two cases determined by the absolute value expression, showing that $f(x)$ equals -1 when $x \leq -\frac{1}{\sqrt{5}}$ and is always greater than or equal to -1 for $x \geq -\frac{1}{\sqrt{5}}$.

12. **Answer:** By the Integer Root Theorem, the possible integer roots are all the divisors of 6 (including negative divisors), so they are $-6, -3, -2, -1, 1, 2, 3, 6$.

Note: correct because the Integer Root Theorem states that any potential integer root

of a polynomial with integer coefficients must be a divisor of the constant term, which in this case is -6, leading to the divisors -6, -3, -2, -1, 1, 2, 3, and 6.

End of Answer Key